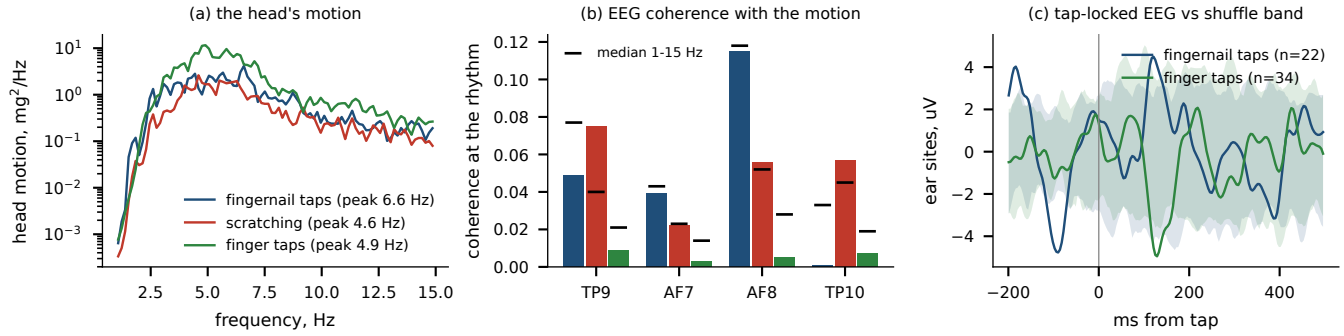
(a) the head's motion
head motion, mg^2/Hz
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
fingernail taps (peak 6.6 Hz)
scratching (peak 4.6 Hz)
finger taps (peak 4.9 Hz)
2.5
5.0
7.5
10.0
12.5
15.0
frequency, Hz
(b) EEG coherence with the motion
coherence at the rhythm
0.12
0.10
0.08
0.06
0.04
0.02
0.00
median 1-15 Hz
TP9
AF7
AF8
TP10
(c) tap-locked EEG vs shuffle band
ear sites, uV
4
2
0
−2
−4
fingernail taps (n=22)
finger taps (n=34)
−200
0
200
400
ms from tap